

5. Write a program to implement facial recognition using neural network.

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import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

# Load dataset (faces)
faces = fetch_olivetti_faces()
X = faces.data          # Features (images flattened)
y = faces.target        # Labels (person IDs)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Neural Network (MLP)
model = MLPClassifier(hidden_layer_sizes=(100,), max_iter=300)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# -----
# Display some predictions
# -----
fig, axes = plt.subplots(2, 5, figsize=(10, 5))
for i, ax in enumerate(axes.flat):
    ax.imshow(X_test[i].reshape(64, 64), cmap='gray')
    ax.set_title(f"Pred: {y_pred[i]}")
    ax.axis('off')

plt.show()

```